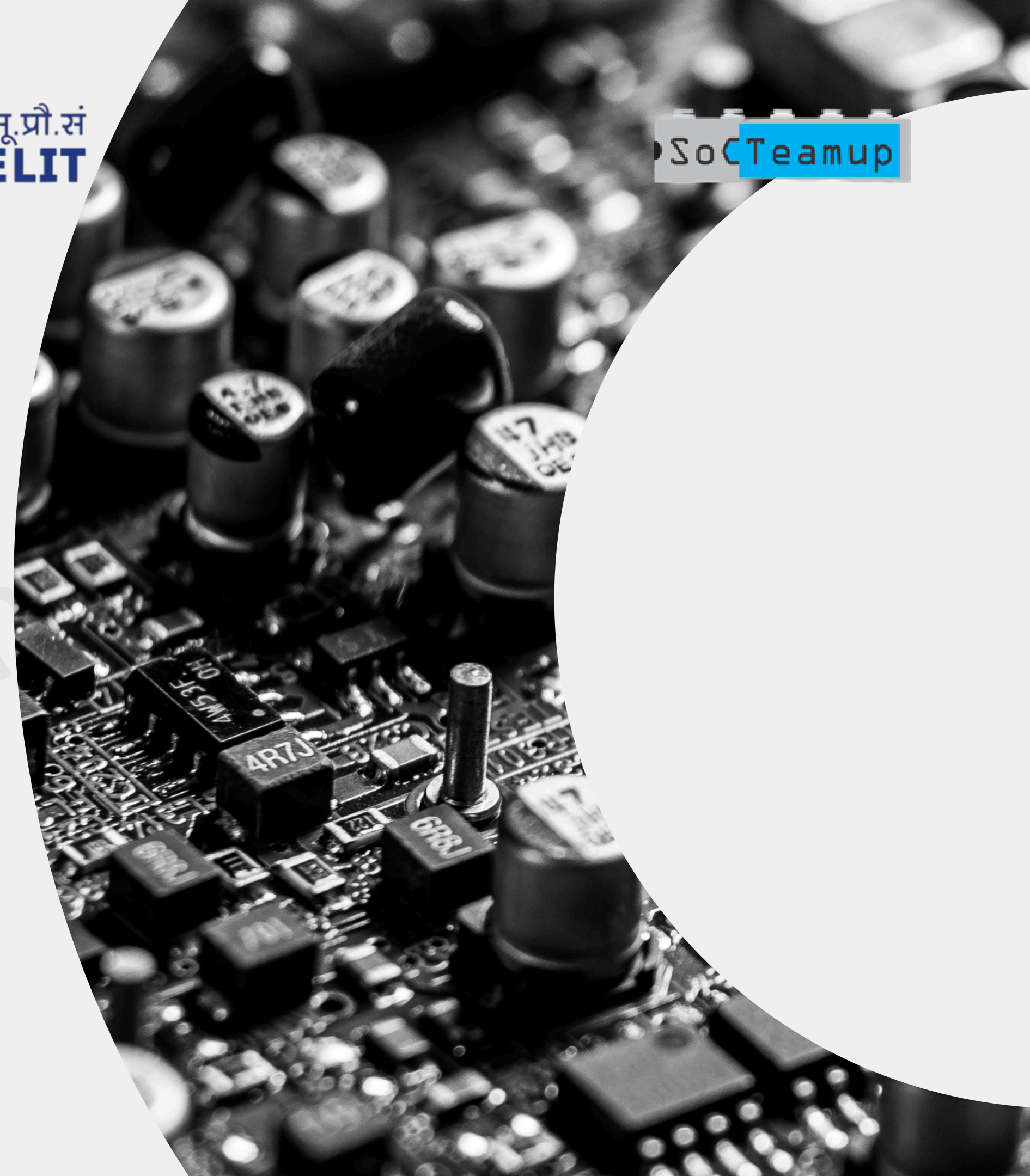


Overview of Design Rule Checking (DRC) and Layout vs Schematic (LVS) checks





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➔ **Introduction: What is Physical Verification?**

After placement and routing, the design must be checked for manufacturability and correctness.

Physical Verification ensures that:

- Layout follows foundry rules (DRC)
- Layout matches the schematic (LVS)
- These checks are mandatory before tapeout (final design submission to the foundry).

Two Key Components:

1. DRC – Design Rule Check → checks geometry & spacing.(min space violation,min Area violation, min width violation)
2. LVS – Layout vs Schematic → checks logical equivalence.

➔ Why DRC and LVS Matter

- Even a small spacing violation or connectivity mismatch can cause:
- Short circuits or opens
- Yield loss during fabrication
- Chip malfunction or failure

DRC and LVS ensure:

- Reliable manufacturing
- Correct electrical behavior
- Compliance with the process design kit (PDK)

➔ Design Rule Checking (DRC): Definition

Design Rule Checking (DRC) verifies that all geometries (wires, vias, transistors, contacts) follow foundry-specified design rules.

Rule Type	Description	Example
Width Rule	Minimum width of a metal layer	$M1 \geq 0.14 \mu m$
Spacing Rule	Minimum spacing between two wires	$M1-M1 \geq 0.14 \mu m$
Enclosure Rule	Via enclosure by metal	Via1 covered by Metal1 and Metal2
Overlap Rule	Diffusion must overlap gate by min area	Poly overlap active $\geq 0.03 \mu m$

Tool Used: Magic (supports real-time DRC visualization)

How DRC Works

- The DRC tool reads the technology rules from the PDK (e.g., SkyWater SKY130 PDK).
- The layout (GDS or MAG) is scanned for rule violations such as spacing, width, and enclosure errors.
- Violations are highlighted visually or reported in a DRC report file.
- Designers fix violations manually or rerun the OpenLane flow with updated parameters.

How DRC Works

DRC Tool Used:

- Magic (called automatically inside OpenLane)

```
Magic576 design.gds  
drc find  
drc count  
drc why
```

Outputs:

- Highlighted errors in layout window
- DRC report file with error types and coordinates

Example: DRC Violation in Magic

- **Scenario:** Metal1 tracks too close
- **DRC Error Message:**
 - DRC error: Metal1 spacing < 0.14 μ m between nets
data[1] and data[2]

- **Fix:**

Move one wire or widen the spacing in Magic:

```
select area  
move e 0.2um
```

Observation:

- The red highlight disappears once spacing meets the rule.

Layout vs Schematic (LVS): Definition

Layout vs Schematic (LVS) ensures that the layout connectivity and devices match the original schematic/netlist.

- **Purpose:**

- Confirms all connections, device types, and counts are identical.
- Detects missing connections, swapped pins, or shorts.

- **Tools Used:**

- Magic → Extract SPICE netlist from layout.
- Netgen → Compares extracted SPICE with schematic/netlist.

LVS Flow Steps

- Extract layout netlist using Magic:

```
Run_magic_spice_export
```

→ produces design.spice

- **Run Netgen comparison:**

```
netgen -batch lvs "design.spice design"
```

```
"synth_netlist.spice design" setup.tcl
```

→ compares layout and schematic connectivity

- **Review LVS report for mismatches.**

Example LVS Output

Expected Output (Pass):

- Circuits match uniquely.
- No unmatched nets or devices found.
- LVS completed successfully.

Error Example:

Net mismatch: Net 'VDD' shorted to 'CLK'

Device mismatch: NMOS count differs (12 vs 13)

Interpretation:

Fix layout or netlist based on report — missing connections or extra devices cause mismatches.

Common **LVS** Errors

Error Type	Description	Fix
Short	Two nets connected unintentionally	Move metal tracks apart
Open	Missing wire or contact	Add missing connection
Parameter mismatch	Device W/L values differ	Adjust layout transistor
Unmatched device	Device present only in one netlist	Check missing layout instance

Note: Each correction must be followed by new extraction and LVS re-run.

Example: Counter Design LVS Flow

- Start with Synthesis Netlist: counter_netlist.v
- Create Layout in Magic: counter.mag
- Extract SPICE Netlist:

```
Run_magic_spice_export
```

- Run_LVS

```
netgen -batch lvs "counter.spice counter"
"counter_netlist.spice counter" setup.tcl
```
- Output: “Circuits match uniquely” if both views are identical.

File Formats in DRC/LVS

File Type	Description	Tool
.tech	Technology file defining DRC rules	Magic
.mag	Layout file	Magic
.spice	Netlist for LVS comparison	Magic, Netgen
.tcl	LVS setup script	Netgen
.rpt	Verification report	Magic, Netgen

Note: The setup file (setup.tcl) maps device names between layout and schematic views.

Common Debugging Tips

Always save a DRC-clean layout before running LVS.

Check power (VDD/VSS) nets first — most LVS mismatches occur here. Run ext2spice with lvs option to avoid parasitic elements during LVS.

Keep consistent net names between schematic and layout.

Re-run DRC after every layout edit — fixing one area can create new violations elsewhere.

➔ Summary and Best Practices

Summary

DRC ensures geometric correctness.

LVS ensures logical equivalence.

Both are essential for signoff before fabrication.

Magic + Netgen provide complete open-source verification capability

Best Practices

Always verify DRC before LVS.

Maintain consistent naming conventions.

Keep your technology files updated (PDK-specific).

Automate checks with simple scripts for faster re-runs.

Document every change between revisions for traceability.

Thank You

echiphu

